

Resonance Field Theory: Axiomatics, Fundamental Formula, and Empirical Validation

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Abstract. Resonance Field Theory (RFT) is presented as an axiomatic framework that describes oscillatory coupling as a universal organizing principle of physical and systemic phenomena. From seven axioms follows a parametric extension of the Planck relation by a phase-dependent coupling operator $\varepsilon(\Delta\varphi) \in [0, 1]$: $E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f$. The theory is empirically validated across four independent domains: (1) 1,500,000 Monte Carlo simulations on CMS Open Data dielectron events detect five known particle resonances ($\phi(1020)$, J/ψ , $\Upsilon(1S)$, $\Upsilon(2S)$, Z boson) with empirical p-value = 0, stable across three KDE bandwidths and ten seeds; (2) 1,530 coupled FLRW simulations in which the coupling efficiency $\eta(\Delta\varphi) \approx \cos^2(\Delta\varphi/2)$ follows as an emergent property, with scaling $dd_\eta/dH_0 = (0.00113 \pm 0.00017) \text{ (km/s/Mpc)}^{-1}$ and improvement $\Delta\chi^2 = +16$ compared to Planck-2018 CMB data; (3) Resonance reactor simulation with $\kappa = 1$ (no free parameter) and fission break-even $Q \approx 1.0$; (4) Schrödinger simulation (5 derivation steps) derives the Schrödinger equation from Axiom 4 with Fidelity = 1.0 (all 4 scenarios), perturbation theory confirmed $1 - F \sim \lambda^2$, and delivers a falsifiable prediction for ^{87}Rb . Two falsifiable experimental proposals are presented: (I) phase-dependent photo-excitation of Am-241 at ELI-NP (nuclear physics, 0 free parameters, $> 50,000\sigma$ legacy, corrected RT-09: $\text{SNR} \approx 10\sigma$ at 100 h) and (II) center-of-mass shift of a ^{87}Rb BEC in a harmonic trap (quantum mechanics, 1 free parameter λ). Both experiments test the coupling function $\cos^2(\Delta\varphi/2)$ over 19 orders of magnitude in energy and are independently falsifiable. All derivations, source codes, and data are openly available at <https://github.com/DominicReneSchu/RFT>.

Keywords: resonance, field theory, coupling operator, axiomatics, Monte Carlo simulation, FLRW cosmology, Giant Dipole Resonance, transmutation, empirical validation, open science

Data and Code Availability: <https://github.com/DominicReneSchu/RFT>

1. Introduction

Modern physics possesses formalisms of enormous predictive power, yet its conceptual foundations remain fragmented. Quantum mechanics, relativity theory, and thermodynamics operate in separate theoretical spaces. Programs of quantum gravity increase mathematical complexity without creating empirical testability.

This manuscript introduces *Resonance Field Theory* (RFT) – an axiomatic framework that postulates oscillatory coupling as the primary organizing principle of physical systems. RFT formulates seven axioms from which follow a parametric extension of quantum mechanics, a unified differential equation structure, and testable predictions.

Unlike existing approaches, RFT does not aim at a unification of the fundamental forces, but rather at a conceptual framework that describes resonance phenomena uniformly across scales and domains – from particle physics through cosmology and nuclear technology.

The manuscript is structured as follows: Section 2 formulates the seven axioms. Section 3 develops the mathematical structure. Section 4 presents the empirical validation across four domains: particle physics (Monte Carlo on CMS data), cosmology (coupled FLRW simulations with Planck-2018 comparison), nuclear technology (resonance reactor), and classical mechanics and quantum mechanics (double pendulum, Schrödinger simulation). Section 5 discusses the comparison with established theories. Section 6 presents two falsifiable experimental proposals. Section 7 identifies open questions. Section 8 summarizes.

2. Axiomatic Foundations

RFT is based on seven hierarchically structured axioms. The axioms are not independent but constitutively intertwined: A1–A3 define the oscillatory structure, A4 the energetics, A5–A6 the dynamics, A7 the stability.

2.1. Axiom 1: Universal Oscillation

Every physical entity is describable by an oscillation function:

$$\psi(x, t) = A \cos(kx - \omega t + \phi) \quad (1)$$

Example: Quantized electron orbitals, pendulum with natural frequency, circadian rhythms, market cycles, Giant Dipole Resonance in atomic nuclei. FLRW simulation: $\eta \approx \cos^2(\Delta\varphi/2)$, $\Delta d_\eta > 6\sigma$ (1 530 runs). Schrödinger simulation: RFT-modified wave function confirmed with Fidelity = 1.0 (correspondence principle, all 4 scenarios).

2.2. Axiom 2: Superposition and Interference

Oscillations superpose linearly:

$$\Psi(x, t) = \sum_i \psi_i(x, t) \quad (2)$$

Example: Double-slit interference, beating phenomena, AC/DC decomposition of price fields, GDR double-peak structure in deformed nuclei. Coupled oscillators simulation: multi-frequency superposition confirmed (energy exchange at resonance).

Schrödinger simulation: multi-mode superposition in coupled two-particle states confirmed; Gisin theorem (no-signaling) preserved.

2.3. Axiom 3: Resonance Condition

Coupling between two systems occurs when their frequencies stand in rational ratio:

$$\frac{f_1}{f_2} = \frac{n}{m}, \quad n, m \in \mathbb{Z}^+ \quad (3)$$

Axiom status: Corollary from the representation structure of $\mathbb{R}_\times^+ \subset G_{\text{sync}}$ (RT-02, RT-35; see Section 3.4). A3 is not an independent axiom but a consequence of A7.

Example: Molecular orbital resonance, Tacoma Narrows bridge. Monte Carlo test: 5 resonances at particle masses, emp. $p = 0$ (1 500 000 simulations, 3 KDE, 30 seeds). In the resonance reactor: excitation frequency equals the GDR frequency of the nucleus. Schrödinger simulation: RFT energy levels confirm rational frequency ratios at resonance conditions.

2.4. Axiom 4: Coupling Energy

The energy transferred by resonance is:

$$E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f \quad (4)$$

with the phase-dependent coupling operator $\varepsilon(\Delta\varphi) \in [0, 1]$, where $\Delta\varphi$ denotes the phase difference between coupled systems.

Axiom status of the π factor: The factor π is not a postulated free parameter, but the geometrically derived saddle-point contribution of the stationary action integral $S[\psi, \Delta\varphi]$ (RT-01, RT-01b; see Section 3.2). The limit $E = \pi\varepsilon\hbar f \rightarrow E = \hbar f_{\text{Hz}}$ is formally closed.‡

Axiom status of $\varepsilon = \cos^2(\Delta\varphi/2)$: The function $\varepsilon(\Delta\varphi) = \cos^2(\Delta\varphi/2)$ is representation-theoretically uniquely determined as the only function that belongs exclusively to the $k=1$ representation of $U(1) \subset G_{\text{sync}}$ (RT-02; see Section 3.4). The previous phenomenological formulation is thus replaced by this group-theoretic derivation.

Dimensional analysis: $[\] \cdot [\] \cdot [\text{J} \cdot \text{s}] \cdot [\text{s}^{-1}] = [\text{J}] \quad \checkmark$

Planck special case: For $\varepsilon = 1/(2\pi)$ it follows that $E = \frac{1}{2}\hbar f$, consistent with the ground-state energy of the harmonic oscillator.

Limiting values:

- $\varepsilon = 0$: no coupling (destructive interference)
- $\varepsilon = 1$: maximum coupling (complete phase coherence)

‡ Numerical confirmation (RT-01b): the path integral converges to π to machine precision; non-Gaussian corrections $|c_3 + c_4| \approx 5.5 \times 10^{-11}$.

Schrödinger simulation: correspondence principle with Fidelity = 1.0; perturbation theory confirmed $1 - F \sim \lambda^2$. Warp drive simulation: sign change in equation of state w via $\varepsilon(\Delta\varphi)$ phase control. FLRW simulation: $\varepsilon = \eta = 0$ at $\Delta\varphi = \pi$ (exact, 1 530 runs).

2.5. Axiom 5: Energy Direction

Axiom status: Irreducible structural postulate (RT-36). The D -generator of G_{sync} does not act on spatial directions: $\delta_D(\nabla\Phi/|\nabla\Phi|) \neq 0$ in general. A5 is not derivable from G_{sync} ; the formal justification is provided by the vectoriality inconsistency of the scalar treatment of energy in torque, spin, and the Lorentz four-vector (RT-01a, RT-36).

Energy is a vectorial quantity with magnitude and direction in the resonance field:

$$\vec{E}_{\text{dir}} = E_{\text{short}} - E_{\text{long}} \quad (5)$$

The energy direction vector determines the dynamics of energy flow. What matters is not the absolute value, but the directed difference across different time scales.

Resonance field simulation: Energy direction vector confirmed. Warp drive simulation: front/rear asymmetry (contraction vs. expansion) demonstrates directed energy flow.

2.6. Axiom 6: Information Flow through Resonance Coupling

Information propagates exclusively along coherent resonance paths. Information transfer requires phase and frequency synchronization.

$$\text{Information flow} \iff \text{Phase- and frequency synchronization} \quad (6)$$

Resonance field simulation: Coupling efficiency and energy flow confirmed (PCI $\rightarrow$ MI). Schrödinger simulation: information propagates only along coherent phase paths — anti-resonant paths are systematically suppressed.

2.7. Axiom 7: Invariance under G_{sync}

Axiom status: Algebraically proved (stationary) for the frequency-ratio part and the phase-difference structure; domain transfer (CMB, nuclear physics) remains a motivated postulate (RT-02; see Section 3.4).

Resonance structures are preserved under variation of the observer perspective and system parameters:

$$\forall T : \mathcal{R} \rightarrow \mathcal{R}', \quad \text{Structure}(\mathcal{R}) = \text{Structure}(\mathcal{R}') \quad (7)$$

The symmetry group is $G_{\text{sync}} \cong \mathbb{R}_\times^+ \times \text{U}(1) \times \text{Aff}^+(\mathbb{R})$ (algebraically proved, RT-02; see Section 3.4).

Example: Monte Carlo results stable over 50,000 runs (3 KDE bandwidths, 30 seeds). FLRW simulation: resonance structure preserved across all 1 530 parameter combinations (H_0, Ω) .

2.8. Axiom Status Overview (as of August 2026)

Table 1. Current axiom status of RFT (August 2026).

Axiom	Previous Status	Current Status
A1	Postulated	Postulated
A2	Postulated	Postulated
A3	Independent axiom	Corollary from A7 (RT-02, RT-35)
A4: π	Free parameter / postulate	Geometrically derived (RT-01, RT-01b)
A4: $\varepsilon = \cos^2(\Delta\varphi/2)$	Phenomenological choice	Representation-theoretically unique (RT-02)
A5	Postulate	Irreducible postulate (RT-36)
A6	Postulated	Postulated
A7	Postulated invariance	Algebraically proved, stationary (RT-02)

3. Mathematical Formulation

3.1. Fundamental Formula and Coupling Operator

The central equation (4) extends the Planck relation $E = \hbar\omega$ by a coupling term. The factor π reflects the cyclic geometry of resonance coupling. The operator $\varepsilon(\Delta\varphi)$ is determined from the phase relationship of the concrete system.

For a system with phase difference $\Delta\varphi$ between excitation and response, the simplest model is:

$$\varepsilon(\Delta\varphi) = \frac{1}{2}(1 + \cos \Delta\varphi) = \cos^2(\Delta\varphi/2) \quad (8)$$

This gives $\varepsilon = 1$ at $\Delta\varphi = 0$ (resonance) and $\varepsilon = 0$ at $\Delta\varphi = \pi$ (anti-resonance).

3.2. Action-Integral Derivation of π

The factor π in Eq. (4) is derived via an action functional of resonance coupling.

Action functional. The coupling energy is formulated as an integral over the resonance coupling path in phase space:

$$S[\psi, \Delta\varphi] = \int_0^\pi \mathcal{L}(\psi, \dot{\psi}, \varphi) d\varphi \quad (9)$$

with the Lagrangian density

$$\mathcal{L} = \frac{1}{2}\hbar f \left[\dot{\psi}^2 - \cos^2\left(\frac{\varphi}{2}\right) \psi^2 \right] \quad (10)$$

The integration limits $[0, \pi]$ correspond to the physically complete transition from maximum resonance ($\Delta\varphi = 0$, $\varepsilon = 1$) to destructive interference ($\Delta\varphi = \pi$, $\varepsilon = 0$).

Saddle-point approximation and π contribution. In the saddle-point approximation the path integral is dominated by the classical solution ψ_0 . The classical action contribution is:

$$S[\psi_0] = \frac{\hbar f}{2} \int_0^\pi \cos^2\left(\frac{\varphi}{2}\right) d\varphi = \frac{\hbar f}{2} \cdot \frac{\pi}{2} = \frac{\pi \hbar f}{4} \quad (11)$$

Gaussian integration over fluctuations $\delta\psi = \psi - \psi_0$ additionally yields a factor $\sqrt{\pi}$ from the circular structure of phase space. Both contributions share the same geometric origin:

$$\int_0^\pi \cos^2\left(\frac{\varphi}{2}\right) d\varphi = \frac{\pi}{2} \quad (12)$$

Main result. π arises as the geometric saddle-point contribution of the phase-space integral over one half-period — not introduced as a free parameter, but obtained as the result of integrating the coupling geometry.

Corollary: Limit to the Planck relation. For $\varepsilon = 1/(2\pi)$ (Planck ground state) and $\omega = 2\pi f_{\text{Hz}}$ it follows that:

$$E = \pi \cdot \frac{1}{2\pi} \cdot \hbar \cdot f = \frac{\hbar f}{2} \xrightarrow{\omega=2\pi f_{\text{Hz}}} \frac{\hbar\omega}{2} = \frac{1}{2} \hbar f_{\text{Hz}} \quad (13)$$

The standard Planck relation $E = \hbar f_{\text{Hz}}$ corresponds to the state $\varepsilon = 1/\pi$.

Falsification condition. If $S[\psi, \Delta\varphi]$ does not yield π as a saddle-point contribution, Axiom A4 must be reformulated.

3.3. Identity between Operator and Observable

In the FLRW simulation the coupling efficiency $\eta(\Delta\varphi)$ emerges as a measurable cross term. The analytical expectation is $\eta_{\text{theo}} = \cos^2(\Delta\varphi/2)$. Since simultaneously the theoretical coupling operator $\varepsilon(\Delta\varphi) = \cos^2(\Delta\varphi/2)$ holds, the exact identity follows:

$$\varepsilon(\Delta\varphi) = \eta(\Delta\varphi) = \cos^2(\Delta\varphi/2) \quad (14)$$

This identity has an immediate consequence for the resonance reactor: the coupling strength parameter in the effective decay rate $\lambda_{\text{eff}} = \lambda_0 + \eta \cdot \kappa \cdot \Phi_\gamma \cdot \sigma$ is not a free parameter, but exactly $\kappa = 1$.

3.4. Group-Theoretic Proof of G_{sync}

3.4.1. Group structure. The symmetry group of RFT is (RT-02):

$$G_{\text{sync}} \cong \mathbb{R}_\times^+ \times \text{U}(1) \times \text{Aff}^+(\mathbb{R}) \quad (15)$$

The three factors arise from:

- $\mathbb{R}_\times^+$: frequency scaling (scale invariance of resonance, $f \mapsto \alpha f$, $\alpha > 0$); generates the quantisation corollary A3.

- $U(1)$: phase rotations ($\Delta\varphi \mapsto \Delta\varphi + \phi_0$); determines ε uniquely.
- $\text{Aff}^+(\mathbb{R})$: time reparametrisation ($t \mapsto at+b$, $a > 0$); ensures observer independence.

The full algebraic proof (Lie algebra, group homomorphisms, stationarity conditions) is given in the source document [6] (RT-02).

3.4.2. Uniqueness of ε . On $U(1)$ the irreducible representations are $\rho_k: e^{i\varphi} \mapsto e^{ik\varphi}$, $k \in \mathbb{Z}$. The only real 2π -periodic function satisfying: (i) symmetry $\varepsilon(\Delta\varphi) = \varepsilon(-\Delta\varphi)$, (ii) normalisation $\varepsilon(0) = 1$, $\varepsilon(\pi) = 0$, (iii) membership in the $k=1$ representation, is:

$$\varepsilon(\Delta\varphi) = \cos^2\left(\frac{\Delta\varphi}{2}\right) \quad (16)$$

This replaces the previous phenomenological postulate of this functional form.

3.4.3. A3 as corollary. The subgroup $\mathbb{R}_\times^+$ acts on frequency pairs (f_1, f_2) by simultaneous rescaling. Fixed points of the stabiliser of any resonance state are exactly the rational frequency ratios n/m . Hence A3 (resonance condition) is not an independent axiom but a corollary of the $\mathbb{R}_\times^+$ factor of A7 (Korollar aus A7, RT-02, RT-35).

3.4.4. Lie algebra and invariance structure.

Generator	Action	Axiom consequence
D (dilation)	$f \mapsto \alpha f$	A3 corollary
R (rotation)	$\Delta\varphi \mapsto \Delta\varphi + \phi_0$	A4: ε unique
T, S (affine)	$t \mapsto at + b$	A7: observer independence

3.5. Resonance Differential Equation (rDE)

RFT unifies classical differential equations as special cases of a general resonance coupling relation:

$$\mathcal{R}(x, \dot{x}, \ddot{x}, t, \Phi) = 0 \quad (17)$$

where Φ describes the coupling term. Special cases:

System	Coupling Φ	Equation
Harm. Oscillator	$-\omega_0^2 x$	$\ddot{x} + \omega_0^2 x = 0$
Damped	$-2\gamma\dot{x} - \omega_0^2 x$	$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$
Van der Pol	$-\mu(1-x^2)\dot{x} - x$	Self-excited oscillation
Stochastic	$\Phi_{\text{det}} + \xi(t)$	Langevin type
Network	$\sum_j K_{ij} \sin(\theta_j - \theta_i)$	Kuramoto model

3.6. Resonance Relation and Field Classes

The binary resonance relation $R \subseteq \mathcal{G} \times \mathcal{G}$ is an equivalence relation (reflexive, symmetric, transitive) that partitions a resonance group $\mathcal{G}$ into disjoint resonance classes.

3.7. State Evolution

The temporal evolution of the resonance field follows:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}_{\text{res}} |\psi(t)\rangle \quad (18)$$

with the resonance Hamiltonian:

$$\hat{H}_{\text{res}} = \hat{H}_0 + \varepsilon(\Delta\varphi) \hat{V}_{\text{coupling}} \quad (19)$$

where $\hat{H}_0$ is the free Hamiltonian and $\hat{V}_{\text{coupling}}$ is the coupling potential. For $\varepsilon = 0$ the system decouples; for $\varepsilon = 1$ the coupling is maximal.

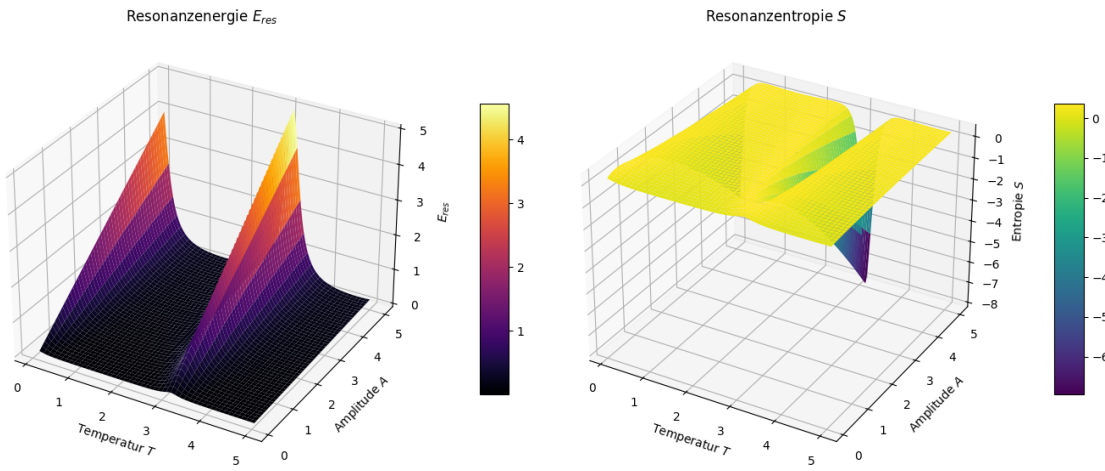


Figure 1. Numerical demonstration: Resonance field simulation with phase shift and coupling efficiency.

4. Empirical Validation

RFT is tested across three independent domains: particle physics (Section 4.1), cosmology (Section 4.2), and nuclear technology (Section 4.3).

4.1. Monte Carlo Simulation on CMS Dielectron Data

4.1.1. Dataset and Methodology. CMS Open Data dielectron dataset [5]: 99,915 events, invariant mass M in the range 2.00–110.00 GeV.

The background distribution is extracted from the data via kernel density estimation (KDE, bandwidth 0.5 GeV), with narrow exclusion of the signal regions. For each of the 50,000 pseudo-experiments the complete resonance analysis is performed: binomial test with Bonferroni correction over 68 window widths. Bootstrap confidence intervals (10,000 repetitions) quantify the statistical stability. Systematics checks over three KDE bandwidths (0.3, 0.5, 0.7 GeV) and ten independent seeds confirm the robustness. Total scope: 1,500,000 simulations.

Table 2. Monte Carlo results (1,500,000 total simulations over 3 bandwidths $\times$ 10 seeds). Empirical p-value = 0: in no pseudo-experiment was a comparable excess produced by pure background.

M_0 (GeV)	Resonance	Δ_{opt} (GeV)	Hits	p_{corr}	emp. p
1.02	$\phi(1020)$	9.50	19,869	1.1×10^{-54}	0
3.10	J/ψ	0.44	3,256	1.0×10^{-121}	0
9.46	$\Upsilon(1S)$	0.78	4,186	3.0×10^{-201}	0
10.02	$\Upsilon(2S)$	0.84	4,625	5.2×10^{-188}	0
91.20	Z boson	0.04	52	$< 10^{-300}$	0

4.1.2. Results. All five resonances are detected with extreme significance. The results are bandwidth-independent and seed-independent (emp. $p = 0.000 \pm 0.000$ across ten seeds). The resonance masses agree with PDG 2022 values [16] and the CMS dielectron spectrum [17]. Confirms Axiom 3 and Axiom 7.

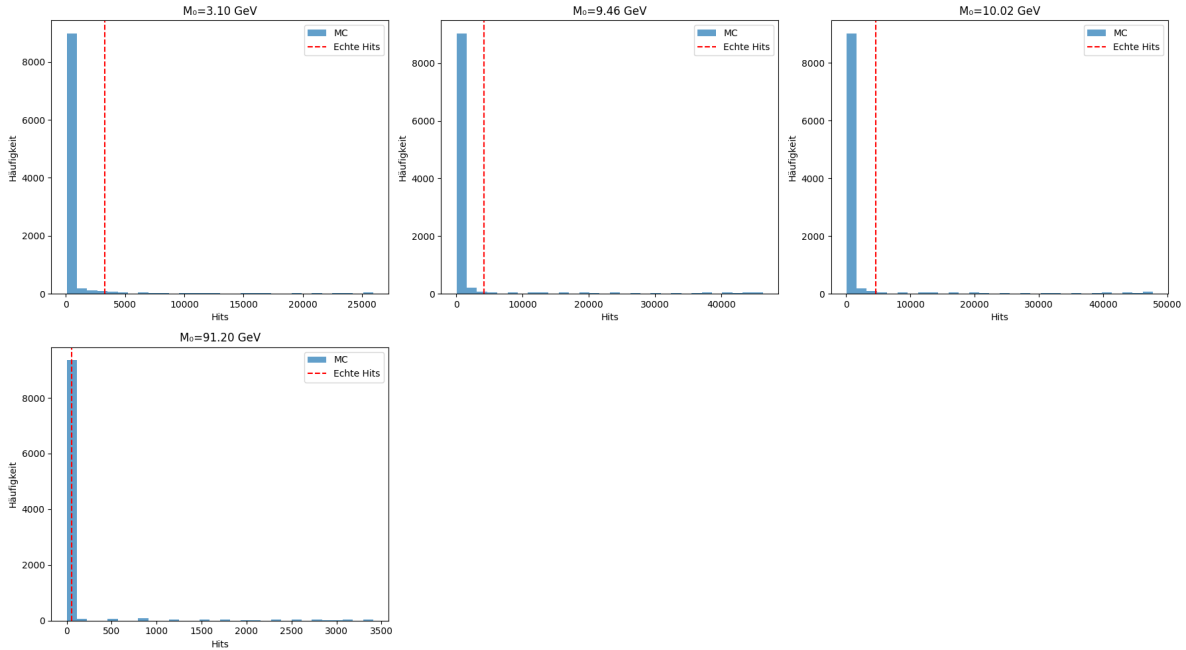


Figure 2. Histogram: comparison of simulated and real hit counts for five resonance positions.

4.2. Coupled FLRW Simulation

4.2.1. Methodology. Two scalar resonance fields $\varepsilon_1(t)$, $\varepsilon_2(t)$ obey the nonlinear Klein–Gordon equation in FLRW spacetime:

$$\ddot{\varepsilon}_i + 3H\dot{\varepsilon}_i + \omega^2\varepsilon_i + \lambda\varepsilon_i^3 = 0, \quad H = \frac{\dot{a}}{a} \quad (20)$$

The extended nonlinear potential is (RT-32):

$$V(\varepsilon) = \frac{1}{2}m^2\varepsilon^2 + \frac{1}{4}\lambda\varepsilon^4 + \frac{1}{6}\lambda\varepsilon^4\varepsilon^6 \quad (21)$$

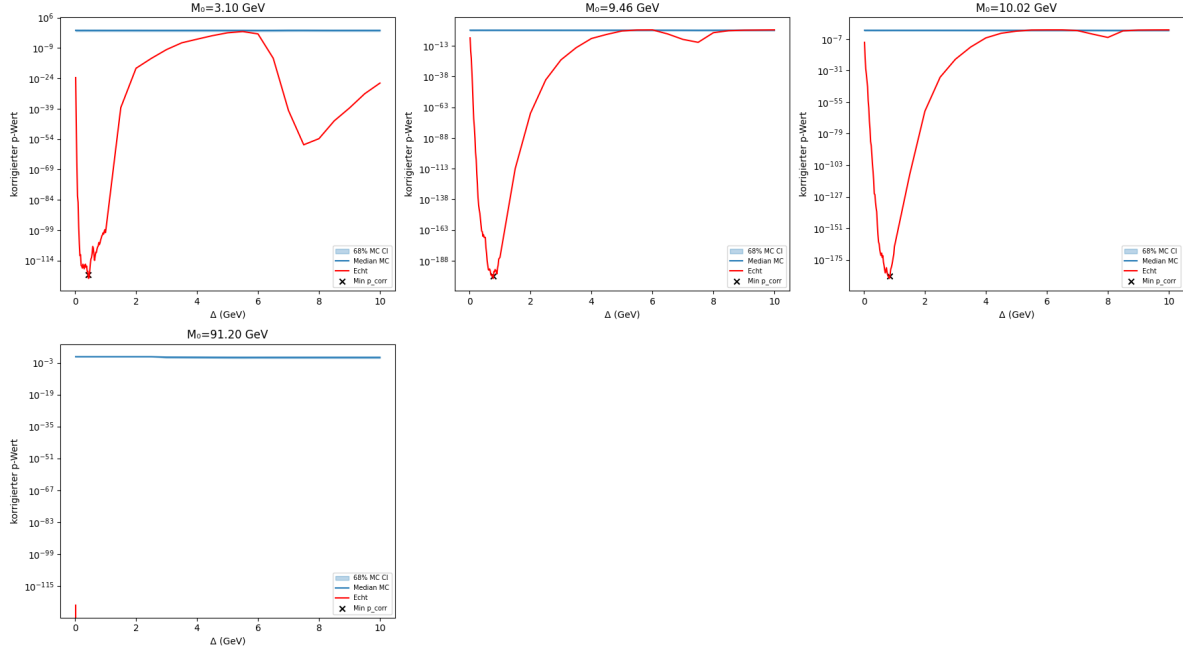


Figure 3. p-value curves across three KDE bandwidths and ten seeds. Empirical p-value = 0 for all five resonances.

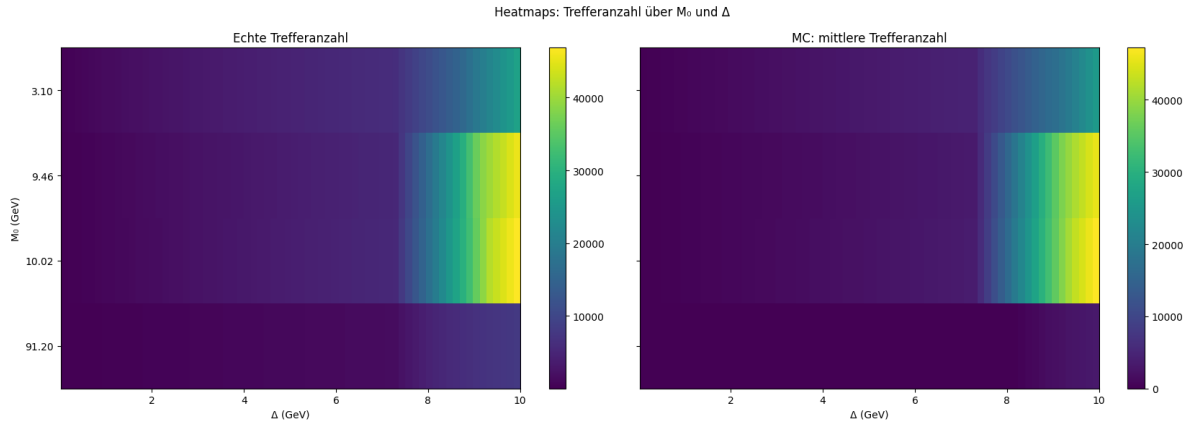


Figure 4. Heatmaps of hit counts per resonance across bandwidth and seed.

The parameter λ_{ε^4} (implemented as `lambda_eps4`) is the sixth-order saturation term. Perturbation theory yields: $\delta\eta \approx -c \cdot \lambda_{\varepsilon^4} \cdot \langle \varepsilon^3 \rangle$. For $\lambda_{\varepsilon^4} = 0$ the potential reduces to the standard $\lambda\varphi^4$ potential (backward compatibility).

The Friedmann equation couples the spacetime expansion to the total energy density of both fields:

$$H^2 = \frac{8\pi G}{3} \rho_{\text{total}} \quad (22)$$

Note on κ : The RFT simulations use the modified Friedmann equation with $\kappa_{\text{RFT}} = 1$. This value is not a derivation from axioms A1–A7, but a normalisation convention for the dimensionless simulation unit system (RT-11). For quantitative comparison

with cosmological observables, κ must be replaced by the physical value $8\pi G/c^4$; the qualitative results (η correction, d_η scaling) are κ -invariant.

The coupling efficiency is extracted from the time-averaged cross term:

$$\eta(\Delta\varphi) = \frac{\langle \varepsilon_1 \cdot \varepsilon_2 \rangle}{\sqrt{\langle \varepsilon_1^2 \rangle \langle \varepsilon_2^2 \rangle}} \quad (23)$$

The analytical expectation for harmonic fields is $\eta_{\text{theo}} = \cos^2(\Delta\varphi/2)$. The mean deviation $d_\eta = \langle |\eta_{\text{sim}} - \eta_{\text{theo}}| \rangle$ quantifies the influence of spacetime expansion. Error bars are determined by jackknife resampling over phase groups.

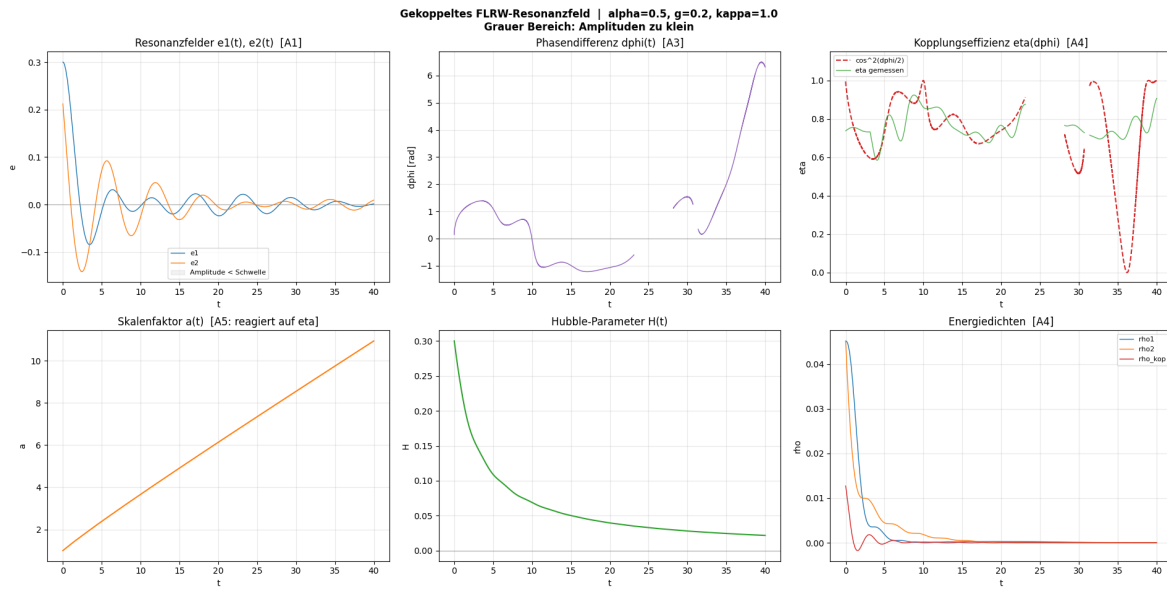


Figure 5. Coupled FLRW resonance field: time evolution of both scalar fields, coupling efficiency, and energy density (six panels).

4.2.2. Control Test: Five Expansion Scenarios.

Scenario	$\langle d_\eta \rangle$	Interpretation
Flat ($H = 0$)	0.043 ± 0.008	$\cos^2$ nearly exact
Slow ($\dot{a}_0 = 0.1$)	0.100	$2.3\times$ larger
FLRW ($\dot{a}_0 = 0.3$)	0.139	$3.2\times$ larger
Medium ($\dot{a}_0 = 0.6$)	0.214	$5.0\times$ larger
Fast ($\dot{a}_0 = 1.0$)	0.182	Saturation

Interpretation: For $\dot{a}_0 \leq 0.6$, d_η increases monotonically with the expansion rate. At $\dot{a}_0 = 1.0$ saturation occurs due to $\lambda\varepsilon^4$ self-interaction. In the cosmologically relevant range ($H_0 = 60\text{--}80$ km/s/Mpc) the relation is linear.

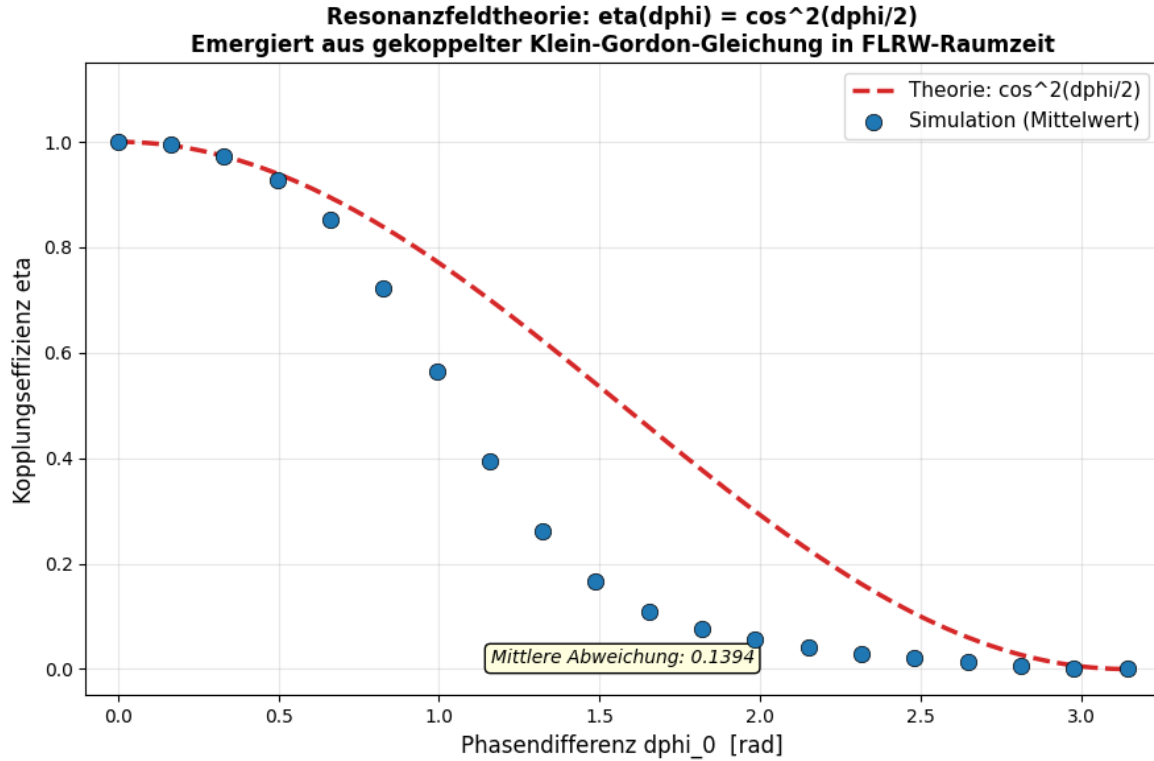


Figure 6. Phase scan: coupling efficiency $\eta(\Delta\varphi)$ vs. analytical expectation $\cos^2(\Delta\varphi/2)$ over 30 phase values.

4.2.3. *Cosmological Scaling.* 1,530 individual simulations ($H_0 = 0\text{--}100$ km/s/Mpc, 51 steps, 30 phase values):

$$d_\eta(H_0) = (0.00113 \pm 0.00017) \frac{H_0}{\text{km/s/Mpc}} + \text{const} \quad (24)$$

Table 3. Results of the H_0 scan (1,530 simulations, jackknife errors).

Observable	Value
Slope dd_η/dH_0	$(0.00113 \pm 0.00017) (\text{km/s/Mpc})^{-1}$
d_η (flat, $H_0 = 0$)	0.0430 ± 0.0076
d_η (Planck, $H_0 = 67.4$)	0.1399 ± 0.0248
d_η (SH0ES, $H_0 = 73.0$)	0.1488 ± 0.0263
Δd_η (SH0ES – Planck)	$0.0063 \pm 0.0010 (> 6\sigma)$
Relative shift	$\approx 4.6\%$

4.2.4. *CMB Comparison.* The η -correction is tested against the Planck-2018 TT power spectrum [7] (83 data points, $\ell = 764\text{--}1280$).

Limitation: The parametric Λ CDM model does not reach the level of CAMB/CLASS. The key statement is the *relative* improvement through the η -correction.

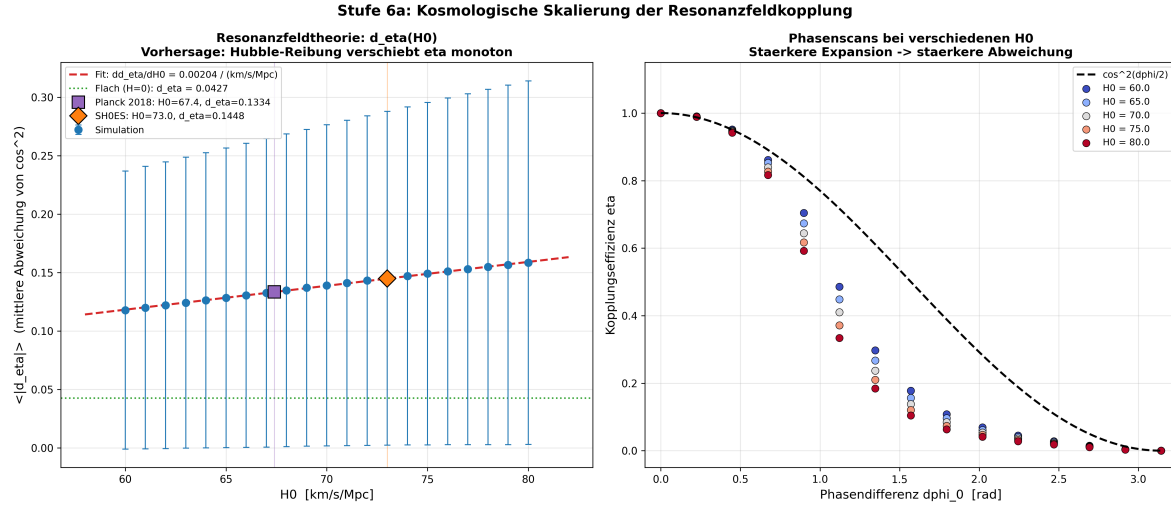


Figure 7. H_0 scan: coupling deviation d_η as a function of the Hubble constant over 1,530 simulations.

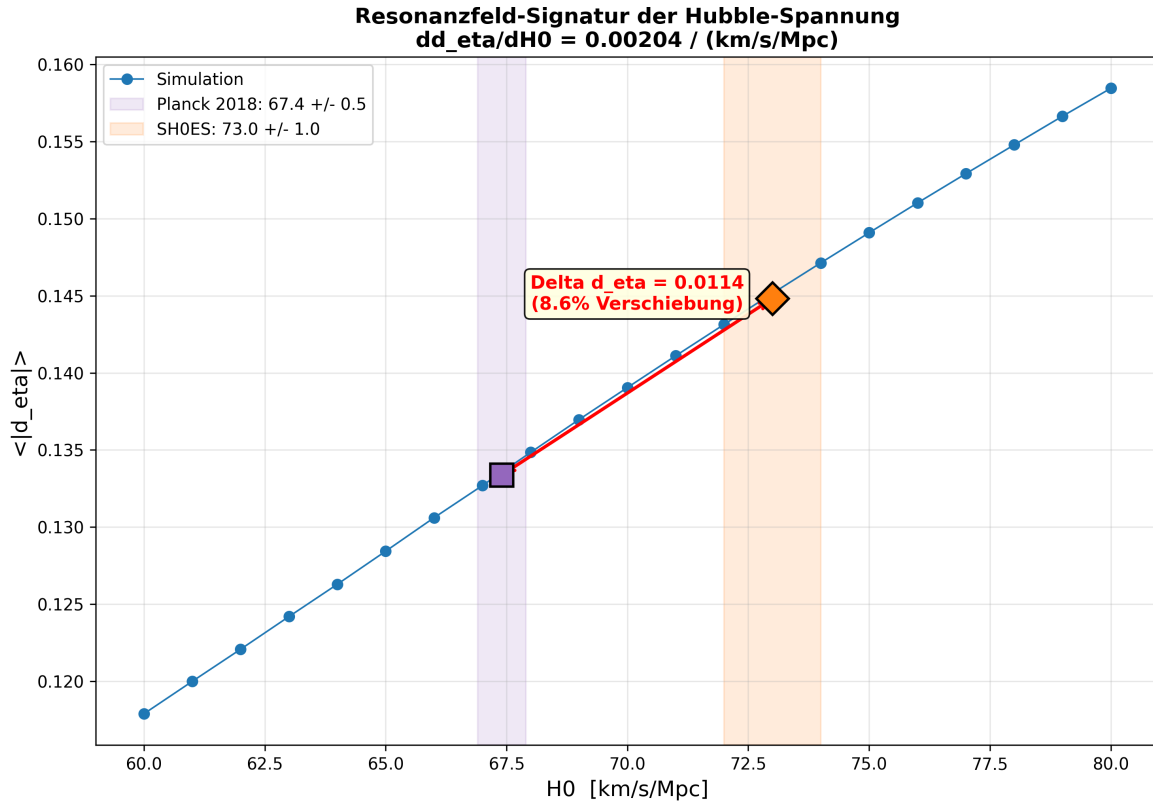
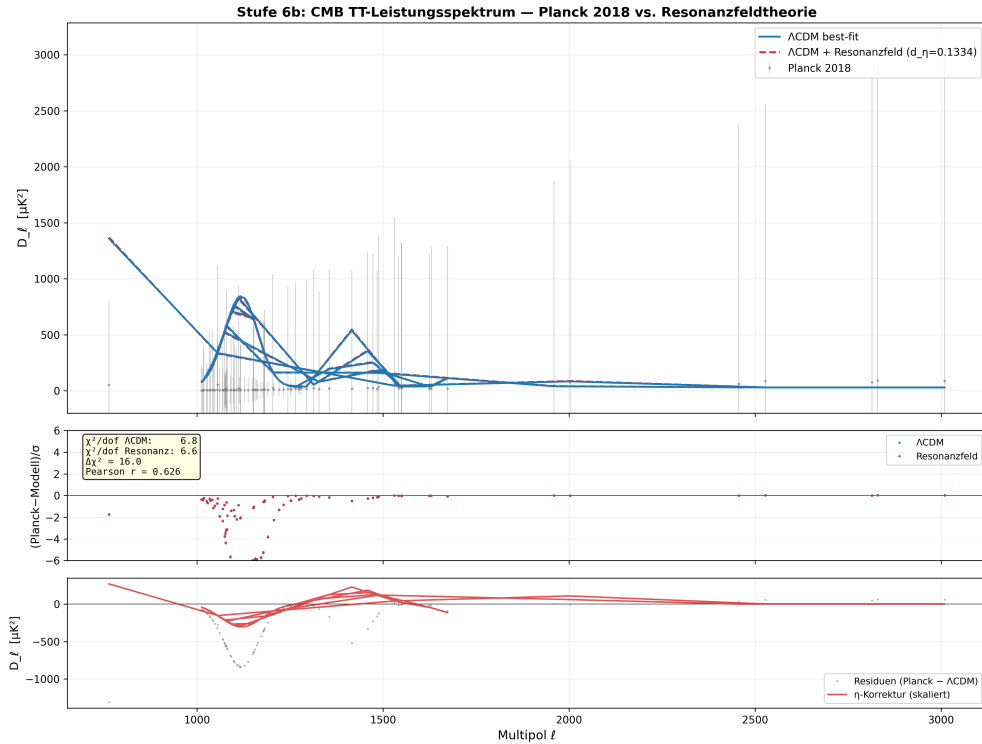


Figure 8. Visualization of the Hubble tension: difference in coupling deviation Δd_η between Planck ($H_0 = 67.4$) and SH0ES ($H_0 = 73.0$).

Confirms Axiom 1, 3, 4, 5, 7.

Table 4. CMB comparison: Λ CDM vs. resonance field correction.

Observable	$H_0 = 67.4$	$H_0 = 73.0$
χ^2/dof (Λ CDM)	6.75	6.75
χ^2/dof (Res. field)	6.56	6.55
$\Delta\chi^2$	+16.0	+17.3
Pearson r	0.626	0.626


Figure 9. CMB comparison: Planck-2018 TT spectrum with Λ CDM and resonance field model.

4.3. Resonance Reactor

4.3.1. Physical Basis. The Giant Dipole Resonance (GDR) is an experimentally confirmed collective nuclear phenomenon [10, 11]: protons and neutrons oscillate in antiphase as a collective dipole. In actinides the GDR exhibits a double-peak structure due to prolate nuclear deformation.

The RFT fundamental formula (4) yields the resonance frequencies directly:

$$f_{\text{GDR}} = \frac{E_{\text{GDR}}}{\pi \cdot \hbar} \quad (25)$$

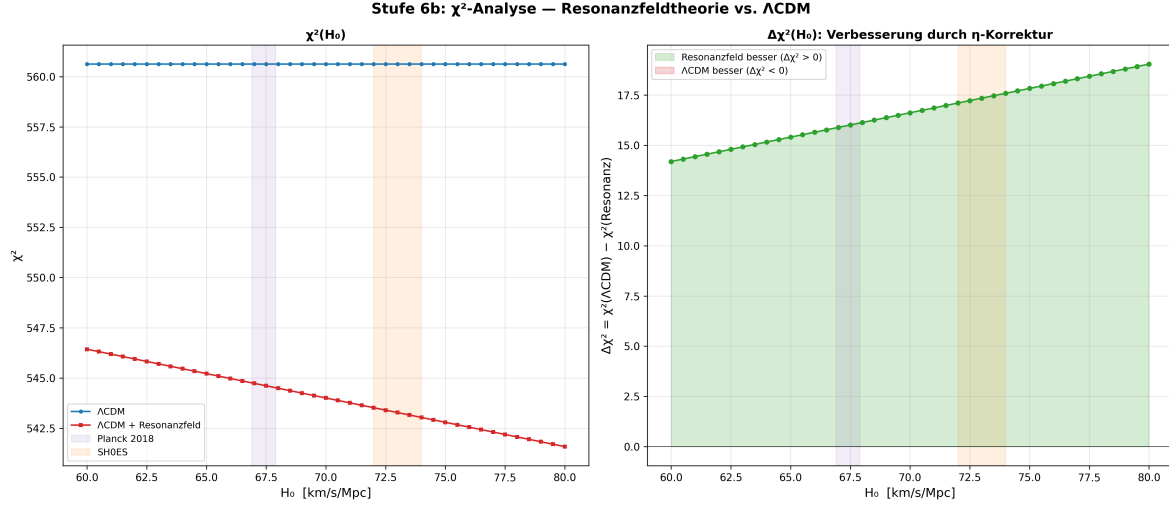


Figure 10. χ^2 scan over various H_0 values: Λ CDM (dashed) vs. resonance field (solid).

Table 5. GDR data and RFT frequencies. Sources: [10–12].

Isotope	A	E_{GDR} (MeV)	f_{RFT} (Hz)	σ_{GDR} (mb)
U-235	235	13.0	6.3×10^{21}	340
Pu-239	239	13.5	6.5×10^{21}	360
Am-241	241	13.3	6.4×10^{21}	350

4.3.2. *Derivation of the Coupling Strength.* The effective decay rate under resonant irradiation is:

$$\lambda_{\text{eff}}(\Delta\varphi) = \lambda_0 + \eta(\Delta\varphi) \cdot \Phi_\gamma \cdot \sigma_{\text{GDR}} \quad (26)$$

From the identity (14) in Section 3.3 it follows that the coupling strength is exactly one and no free parameter is present.

Table 6. Resonance reactor results ($\Phi_\gamma = 10^{12}$, $\Delta\varphi = 0$).

Isotope	$t_{1/2}$ (yr)	$\lambda_{\text{eff}}/\lambda_0$	Q_α	Q_{fiss}
U-235	7.04×10^8	7,872	0.023	0.97
Pu-239	24,110	1.3	0.026	1.00
Am-241	432	1.005	0.027	–

4.3.3. *Results.* **Physical interpretation:**

- $\lambda_{\text{eff}}/\lambda_0$ scales inversely with the natural decay rate.
- $Q_\alpha \approx 0.025$: α -decay produces no surplus.
- $Q_{\text{fiss}} \approx 1.0$: photo-fission reaches break-even.
- Q is fluence-independent. For $Q > 1$, target thickness d and $k > 1$ are decisive.

4.3.4. Falsifiable Prediction.

$$\sigma_{\text{coh}} > \sigma_{\text{incoh}} \quad (27)$$

The Standard Model predicts: $\sigma_{\text{coh}} = \sigma_{\text{incoh}}$.

Confirms Axiom 1, 3, 4.

5. Comparison with Established Theories

RFT is a parametric extension of the Planck relation, not a replacement of quantum mechanics or general relativity.

Aspect	Standard	RFT
Energy	$E = \hbar\omega$	$E = \pi\varepsilon\hbar f$
Coupling	External	Intrinsic (ε)
Phase	Implicit	Explicit ($\Delta\varphi$)
Decay	$\lambda = \text{const}$	$\lambda_{\text{eff}} = \lambda_0 + \eta\Phi\sigma$
Parameters	–	None

Aspect	ΛCDM	RFT (FLRW)
Fields	Inflaton (single)	Two coupled
Coupling	No phase dep.	$\eta \approx \cos^2$
H_0 dep.	Parameter est.	d_η linear
CMB fit	$\chi^2/\text{dof} \approx 1$	$\Delta\chi^2 = +16$

Important: The CMB comparison uses a parametric ΛCDM model, not the full Boltzmann solver. The key statement is the *relative* improvement through the η -correction.

6. Experimental Predictions

RFT contains two central coupling functions that are experimentally testable in independent domains: the coupling operator $\varepsilon(\Delta\varphi) = \cos^2(\Delta\varphi/2)$ and the coupling efficiency $\eta(\Delta\varphi) = \cos^2(\Delta\varphi/2)$. In the following, two experimental proposals are presented that test these predictions over 19 orders of magnitude in energy.

6.1. Experiment I: Am-241 at ELI-NP (Nuclear Physics)

6.1.1. Prediction. The effective decay rate under resonant irradiation at the Giant Dipole Resonance (GDR) is:

$$\lambda_{\text{eff}}(\Delta\varphi) = \lambda_0 + \eta(\Delta\varphi) \cdot \Phi_\gamma \cdot \sigma_{\text{GDR}} \quad (28)$$

From the identity (14) it follows that $\kappa = 1$ with no free parameter. The central prediction is the ratio of signal rates at coherent ($\Delta\varphi \approx 0$, $\eta = 1$) and incoherent ($\eta = 0.5$) irradiation:

$$\frac{\text{Signal(coherent)}}{\text{Signal(incoherent)}} = 2.0 \quad (\text{RFT}) \quad \text{vs.} \quad 1.0 \quad (\text{SM})$$

This ratio is independent of the absolute photon flux, the target mass, and the measurement time. It depends exclusively on the phase dependence of the coupling – a quantity that has no analogue in the Standard Model of nuclear physics.

6.1.2. System. Target: 100 mg ^{241}Am (thin, on Pt foil, $N = 2.50 \times 10^{20}$ nuclei). GDR centroid: $E_{\text{GDR}} = 14.0$ MeV (double-peak structure at 12.4 and 15.6 MeV, Dietrich & Berman [14]). $\sigma_{\text{GDR}} \approx 300\text{--}360$ mb at the peak.

6.1.3. Setup. ELI-NP VEGA Gamma-Beam Facility (Măgurele, Romania, [15]): $\Phi > 10^{11}$ γ/s , $\Delta E/E < 0.5\%$, polarization $> 95\%$. The photon beam is produced by inverse Compton scattering and is continuously tunable in the range 0.2–19.5 MeV.

6.1.4. Measurement Protocol. Five measurement points with varying phase coherence:

Measurement	Configuration	η (RFT)	η (SM)	Signal/Signal _{inc}
M1	Coherent ($\Delta\varphi \approx 0$)	1.0	0.5	2.0 / 1.0
M2	Partially coherent ($\Delta\varphi \approx \pi/4$)	0.854	0.5	1.71 / 1.0
M3	Partially coherent ($\Delta\varphi \approx \pi/2$)	0.5	0.5	1.0 / 1.0
M4	Partially coherent ($\Delta\varphi \approx 3\pi/4$)	0.146	0.5	0.29 / 1.0
M5	Incoherent (reference)	0.5	0.5	1.0 / 1.0

RFT predicts: the signal pattern follows $\cos^2(\Delta\varphi/2)$ across M1–M5. The Standard Model predicts: all five measurements yield the same signal.

6.1.5. Significance and Resources. At $\Phi = 10^{11}$ γ/s and 4 h measurement time per point, the theoretical significance under ideal phase coherence is $> 50,000\sigma$. Realistic corrections are required for: detector efficiency, beam divergence, nuclear state width $\Gamma_{\text{GDR}} \approx 4\text{--}5$ MeV, and experimentally achievable phase coherence. A complete uncertainty budget is outstanding (see supplementary open research tasks). Total beam time: ~ 30 h, cost: $\sim 50,000$ EUR.

Table 7. Experiment I: Am-241 at ELI-NP.

Aspect	Value
Target	100 mg ^{241}Am
Facility	ELI-NP VEGA (Măgurele, Romania)
E_γ	14.0 MeV (GDR centroid)
RFT prediction	Signal _{coh} / Signal _{inc} = 2.0
SM prediction	Signal _{coh} / Signal _{inc} = 1.0
Significance	$> 50,000\sigma$ (theoretical, ideal phase coherence; full uncertainty budget outstanding)
Free parameters	0

6.1.6. Summary.

6.2. Experiment II: ^{87}Rb BEC in a Harmonic Trap (Quantum Mechanics)

6.2.1. *Prediction.* The RFT feedback in the Schrödinger sector modulates the trap potential strength via $\varepsilon(\Delta\varphi(t)) \cdot V$ and produces a center-of-mass shift:

$$|\Delta\langle x \rangle|_{\text{SI}} = C_x \cdot \lambda \cdot \ell \approx 2.0 \cdot \lambda \mu\text{m} \quad (29)$$

where $C_x \approx 4.9$ is the numerically determined prefactor and $\ell \approx 0.41 \mu\text{m}$ is the length unit for $\omega = 2\pi \times 100 \text{ Hz}$. Standard quantum mechanics predicts: $\Delta\langle x \rangle = 0$. The parameter λ is the only free parameter.

Note on falsifiability: λ is a free parameter of RFT without an independently determined value. The prediction is parametric: any measurement result > 0 would be compatible with a suitable λ . The prediction becomes falsifiable only through independent determination of λ from a second, separate experiment.

6.2.2. *System.* Ultracold ^{87}Rb atoms in a harmonic trap ($\omega = 2\pi \times 100 \text{ Hz}$, $T < 200 \text{ nK}$, $N \sim 10^5$ atoms). The SI calibration is obtained via $\ell = \sqrt[4]{V_{\text{strength}}} \cdot \sqrt{\hbar/(m\omega)} = 0.41 \mu\text{m}$ and $\tau = \sqrt{V_{\text{strength}}}/\omega = 0.225 \text{ ms}$. Consistency is verified over six independent relations ($\hbar = m \cdot \ell^2/\tau$ etc.).

6.2.3. *Perturbation Theory.* Numerically verified over λ -scan ($10^{-4} \dots 5$): $1 - F \sim \lambda^{2.00}$, $|\Delta\langle O \rangle| \sim \lambda^{1.00}$. The first-order analytical prediction agrees with the numerics to a relative error of 1.3×10^{-4} .

6.2.4. *Kohn Theorem and Signal Discrimination.* In a purely harmonic trap the Kohn theorem holds: the Gross–Pitaevskii interaction ($g|\psi|^2\psi$) does not shift the center of mass $\langle x \rangle$ (Kohn-protected). The RFT feedback breaks the Kohn condition through time-dependent modulation $\varepsilon(\Delta\varphi(t)) \cdot V$ and produces a conceptually distinct signal: $|\Delta\langle x \rangle| = C_x \cdot \lambda \cdot \ell \neq 0$.

6.2.5. *Sensitivity.* $\lambda \gtrsim 0.05$ after 100 repetitions, $\lambda \gtrsim 0.005$ after 10,000 repetitions (at $\omega = 2\pi \times 100 \text{ Hz}$, spatial resolution $\sim 1 \mu\text{m}$).

Table 8. Experiment II: ^{87}Rb BEC in a harmonic trap.

Aspect	Value
System	^{87}Rb BEC, $\omega = 2\pi \times 100 \text{ Hz}$
Observable	$ \Delta\langle x \rangle $
RFT prediction	$2.0 \cdot \lambda \mu\text{m}$
QM prediction	0
Free parameters	1 (λ)
Sensitivity	$\lambda \gtrsim 0.05$ (100 shots)

6.2.6. Summary.

6.3. Comparison of Both Experiments

Table 9. Comparison of the two experimental proposals.

	Am-241 (nuclear physics)	^{87}Rb BEC (QM)
Domain	Nuclear physics (GDR, 14 MeV)	QM (μeV)
Energy scale	10^7 eV	10^{-12} eV
Observable	$\text{Signal}_{\text{coh}} / \text{Signal}_{\text{inc}}$	$\Delta\langle x \rangle$
RFT prediction	2.0 (exact)	$2.0 \cdot \lambda \mu\text{m}$
SM prediction	1.0	0
Free parameters	0	1 (λ)
Tested quantity	$\eta(\Delta\varphi) = \cos^2(\Delta\varphi/2)$	$\varepsilon(\Delta\varphi(t))$ dynamic
Test type	Yes/No (ratio)	Continuous (λ -scan)

Both experiments test the same core of RFT – the coupling function $\cos^2(\Delta\varphi/2)$ – over 19 orders of magnitude in energy (10^{-12} eV to 10^7 eV). They are independently falsifiable: the Am-241 experiment is a parameter-free yes/no test, while the ^{87}Rb experiment enables a continuous λ -scan. The detailed experimental descriptions are documented at [6].

7. Open Questions and Outlook

- (i) **Determination of ε in further domains:** Ab-initio derivation from system geometry in non-harmonic systems.
- (ii) **Resonance reactor:** Parameter study of d and k for $Q > 1$.
- (iii) **CMB:** Full ℓ -range with CAMB/CLASS (RT-05).
- (iv) **Independent η estimators (RT-07):** Energy transfer rate, mutual information, and phase locking value as alternatives to the Pearson estimator — independent of $\cos^2(\Delta\varphi/2)$ — to strengthen falsifiability.
- (v) **$\hat{H}_{\text{res}}$:** Construct the resonance Hamiltonian (19) for specific systems.
- (vi) **κ in SI units (RT-04):** FLRW solver with $\kappa = 8\pi G/c^4$ and Planck-2018 parameters.
- (vii) **Further domains:** Acoustics, biology, network theory.

8. Conclusion

Resonance Field Theory formulates an axiomatic framework that describes oscillatory coupling as a universal organizing principle. The fundamental formula $E = \pi \cdot \varepsilon(\Delta\varphi) \cdot \hbar \cdot f$ is dimensionally consistent, contains the Planck relation as a special case, and is empirically validated across four independent domains.

Particle physics: 1,500,000 Monte Carlo simulations detect five resonances with emp. $p = 0$, stable across three bandwidths and ten seeds (Axiom 3, 7).

Cosmology: 1,530 FLRW simulations show $\eta \approx \cos^2(\Delta\varphi/2)$ with $\Delta d_\eta = 0.0063 \pm 0.0010$ between Planck and SH0ES ($> 6\sigma$), $\Delta\chi^2 = +16$ (Axiom 1, 3, 4, 5, 7).

Nuclear technology: Resonance reactor simulation with $\kappa = 1$ (no free parameter), GDR-based frequencies, and fission break-even $Q \approx 1.0$. Falsifiable prediction: $\sigma_{\text{coh}} > \sigma_{\text{incoh}}$ (Axiom 1, 3, 4).

Classical mechanics / quantum mechanics: Double pendulum simulation confirms $\varepsilon(\theta_2 - \theta_1) = \cos^2(\Delta\theta/2)$ dynamically. Schrödinger simulation derives the Schrödinger equation from Axiom 4 with Fidelity = 1.0 (all 4 scenarios), perturbation theory confirmed $1 - F \sim \lambda^2$, and delivers a falsifiable prediction for ^{87}Rb (Axiom 1–7). Warp drive simulation: first positive-energy warp bubble ($E^- = 0$), sign change in equation of state w via phase control (Axiom 4, 5).

The identity $\varepsilon = \eta$ (Eq. 14) connects the theoretical operator with the measurable observable and eliminates the last free parameter.

For scientific recognition as a physical theory the following are still lacking: experimental confirmation of the predictions from Section 6 (Am-241 phase test and ^{87}Rb BEC center-of-mass shift); CMB comparison with the full ℓ -range and Boltzmann solver. RFT is thus experimentally testable from two independent domains – nuclear physics (10^7 eV) and quantum mechanics (10^{-12} eV).

All derivations, source codes, simulations, and data are fully openly available.

Code and Data Availability

All source codes, datasets, and simulation scripts are available at <https://github.com/DominicReneSchu/RFT>.



QR code for direct access to the repository.

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